Abstract:

Alzheimer's Disease (AD) is a neurodegenerative disorder commonly found among adults 65 years and older. Two-thirds of AD patients in the United States are females. However, analysis of publicly available data show minimal sexual disparity identified using the cognitive questionnaires known as the Mini-Mental State Examination, Activities of Daily Life, and Functional Assessment Test. In contrast, sexual disparity revealed pronounced observations in mice with an AD-like phenotype using a spatial navigation test. When rTg4510 mice (overexpressing tau protein and a brain atrophy indicative of AD) were assessed on the Morris Water Maze, the rTg4510 mice showed reduced performance in the distance traveled, velocity swam, and time taken to escape. Moreover, there was an increased clinging to walls. Further analysis between male and female mice showed females performing worse in the distance traveled and velocity swam than males.

Analysis of an RNA Seq dataset comparing wild type and rTg4510 mice for fifty of the most commonly associated genes with spatial navigation identified 26 genes differentially regulated. Of these, only one (Plcb3) showed a significant difference in expression between males and females. In both rTg4510 mice and AD humans, Plcb3 is upregulated. Furthermore, there is a higher level of expression of the Plcb3 gene in female rTg4510 mice. Plcb3 regulates calcium signaling for spatial memory. Higher levels of Plcb3 may cause an excessive calcium overload to occur. Human studies have demonstrated that AD affects allocentric navigation, however, sex-specific differences have not been well explored within this area of research.